

Workplace Respiratory Infection Policy Analysis

A personal portfolio project using Quasi-Poisson regression to model infection counts with worked-hours exposure.

Executive Summary

This project evaluates aggregated respiratory infection records from a hospital network to identify which operational factors are associated with lower infection rates. The analysis starts with a Poisson count model and uses $\log(\text{worked hours})$ as an offset, then switches to a Quasi-Poisson model after detecting substantial overdispersion.

The final model indicates that staff experience is the strongest predictor of infection risk. PPE regime, wage-increase coverage, and external training coverage do not add meaningful explanatory value once experience is included.

Dataset Snapshot

Metric	Value
Aggregated records	72
Recorded infections	7,717
Worked hours	397,350
Overall infection rate	19.4 per 1,000 hours

Model Choice

The initial Poisson model produced a dispersion ratio of 49.4, far above the overdispersion threshold of 1.5. A standard Poisson model would understate uncertainty, so the project uses a Quasi-Poisson model for inference.

Final Model Results

Term	IRR	95% CI	Interpretation
Experience level 2	0.421	0.272 to 0.650	Approx. 58% lower rate than Level 1
Experience level 3	0.434	0.297 to 0.633	Approx. 57% lower rate than Level 1
Experience level 4	0.254	0.155 to 0.417	Approx. 75% lower rate than Level 1

Experience	Infections	Worked hours	Rate per 1,000 hours
Level 1	2,460	59,528	41.3
Level 2	1,580	90,865	17.4
Level 3	2,620	146,220	17.9
Level 4	1,057	100,737	10.5

Professional Interpretation

The analysis suggests that experienced staff function as an infection-control asset. Compared with Level 1 staff, Level 4 staff are associated with an estimated 75% lower infection rate. This is a stronger and more stable signal than the PPE-regime differences visible in the raw exploratory plots.

A reasonable operational response is to focus on retention, mentoring, and knowledge transfer. PPE policy should still be maintained and audited, but this dataset does not support choosing a single PPE regime based on raw infection counts alone.

Recommendations

- Prioritise retention of experienced nursing staff as part of infection-control planning.
- Pair less experienced staff with senior mentors in higher-risk units.
- Review PPE standards as a system-level control rather than selecting a regime from unadjusted rates.
- Reassess external training content and measurement, since training coverage alone does not explain lower infection rates here.

Limitations

The data is observational and aggregated, so the results should not be interpreted as individual-level causality. Unmeasured factors such as patient acuity, local outbreak pressure, ward type, shift pattern, and staffing mix may also influence infection rates.

Portfolio Notes

This PDF is the polished, employer-facing version of the analysis. The repository preserves the original source exports in an archive folder for provenance, while the main README, analysis source, and report are written as a personal statistical modelling project.